

RSTSGUI2 - Raman & Thomson Scattering Analysis User Manual

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Version History

V9.9 (October 2024) - RSTSGUI2

V9.8 (Prior)

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Overview

RSTSGUI2 is a MATLAB-based graphical user interface for analyzing Raman Scattering (RS) and Thomson Scattering (TS) spectroscopy data from plasma diagnostics. The tool provides:

- **Automated intensity calibration** using Raman scattering from known gases
- **Multiple fitting algorithms** for electron temperature (T_e) and density (n_e)
- **Realistic error estimation** using Monte Carlo sampling
- **Interactive parameter adjustment** and visualization

Key Features (V9.9)

- ☐ Weighted fitting prioritizes central spectral region (532 nm)
 - ☐ Monte Carlo error estimation (~15-20% realistic errors)
 - ☐ Three independent fitting methods for cross-validation
 - ☐ Real-time diagnostic output and parameter adjustment
-

Installation

Requirements

- **MATLAB R2018b or newer** (requires uifigure support)
- **Image Processing Toolbox**
- **Optimization Toolbox**

Setup

1. **Download RSTSGUI2.m** from GitHub:

<https://github.com/ShuraBura/Raman-Thomson-Scattering-GUI>
Branch: `claude/matlab-support-011CUNPs1aDJHFFbyb5mLPrh`

2. **Place in MATLAB path** or navigate to the directory:

```
cd /path/to/Raman-Thomson-Scattering-GUI
```

3. **Edit default file paths** (lines 26-107) to point to your data
4. **Run the GUI:**

Quick Start

Typical Analysis Workflow

1. **Load data** → GUI opens with initial Gaussian fit displayed
2. **Adjust ROI/Exclusion** → Set signal region and laser line exclusion
3. **Recalibrate Raman** → Optimize wavelength calibration (optional)
4. **Choose fitting method:**
 - **Int.Calib.Full** (recommended) → Best absolute ne
 - **Shape Calib.** → Verification/consistency check
 - **Int.Calib.Gauss** → Fast, simplified analysis
5. **Review results** → Check fit quality (R^2 , SSR, visual inspection)
6. **Save outputs** → Export CSV data and plots

GUI Layout

Display Panels (Top)

Raman Image	Thomson Image	File List
Raman Spectrum	Thomson Spectrum Fit	

- **Raman Image:** Raw Raman scattering image with ROI overlay
- **Thomson Image:** Thomson scattering image with ROI overlay
- **Raman Spectrum:** Experimental vs. theoretical Raman spectrum
- **Thomson Spectrum:** Fitted spectrum with current parameters
- **File List:** Browse Thomson files in current directory

Control Panel (Bottom)

Row 1: Image Parameters

- **ROI yc, wy:** Vertical center and width of signal integration region
- **Excl: start to end:** Pixel columns to exclude (laser line)
- **Refresh:** Update images and ROI overlays

Row 2: Acquisition Parameters

- **RS Shots, TS Shots:** Number of laser shots averaged
- **a, b:** Wavelength calibration ($\lambda = a \cdot \text{pixel} + b$)
- **Toggle TS sym:** Switch between line and symbol plot styles

Row 3: Energy & Calibration

- **RS E(mJ), TS E(mJ):** Laser energy per shot (millijoules)
- **Linewidth:** Raman spectral linewidth (nm)
- **P(Pa):** Gas pressure (Pascals)

Row 4: Gas Composition

- **T(K):** Gas temperature (Kelvin)
- **N2, O2:** Gas mixing ratios (must sum to 1.0)
- **Recalib disp. from Raman:** Re-optimize wavelength calibration

Row 5: Filters

- **RmOut:** Remove outliers using IQR method
- **MedF:** Apply median filter to spectrum
- **Window:** Median filter window size
- **Recalc Raman Spectrum:** Update Raman analysis

Row 6: Plasma Parameters

- **Te(eV):** Electron temperature (electron volts)
- **ne:** Electron density (m^{-3})
- **α :** Plasma screening parameter (read-only)
- **γ :** Duty cycle factor (for pulsed plasmas)

Row 7: Fitting Controls

- **Multi:** Number of multi-start optimizations (1-20)
- **Recalc TS:** Manually update spectrum from Te/ne/ α / γ
- **Shape Calib.:** Fit shape to get Te/ α /ne
- **Int.Calib.Gauss:** Gaussian fit with intensity calibration
- **Int.Calib.Full:** Full coherent fit with intensity calibration
- **< / >:** Pixel shift controls for manual alignment

Utilities Panel (Right)

- **Decouple:** Open plots in separate figure windows
 - **EEDF:** Plot electron energy distribution function
 - **Save CSV:** Export wavelength, experimental, and fitted data
 - **Load buttons:** Select new Raman/Thomson files
 - **BG buttons:** Load background images for subtraction
-

Theory & Equations

Raman Scattering (Intensity Calibration)

Raman scattering from known gases (N_2 , O_2) provides absolute intensity calibration.

Calibration Factor:

$$C = I_{\text{exp}} / (E_{\text{total}} \times I_{\text{theory}})$$

Where: - I_{exp} = $\int$ experimental Raman spectrum $d\lambda$ (measured counts) - E_{total} = laser energy $\times$ shots (Joules) - I_{theory} = $\int$ theoretical Raman spectrum $d\lambda$ (calculated from cross-sections)

Theoretical Raman Spectrum:

$$I_{\text{Raman}}(\lambda) = \sum_{\text{species}} \sigma_{\text{species}}(\lambda) \times n_{\text{species}} \times G(\lambda, \text{FWHM})$$

- $\sigma_{\text{species}}(\lambda)$: Raman cross-section for each species (N_2 , O_2)
- n_{species} : Number density of species
- $G(\lambda, \text{FWHM})$: Instrumental broadening (Gaussian)

Thomson Scattering (Plasma Diagnostics)

Thomson scattering measures electron temperature and density from spectral shape and intensity.

Electron Temperature from Spectral Width

FWHM-Te Relation (for Gaussian approximation):

$$T_e = (m_e c^2) / (32 \ln 2) \times (FWHM / \lambda_{\text{laser}})^2$$

Where: - $m_e c^2 = 0.511 \text{ MeV}$ (electron rest energy) - FWHM: Full width at half maximum (nm) - $\lambda_{\text{laser}} = 532 \text{ nm}$ (laser wavelength)

Physical Interpretation: Wider spectrum $\rightarrow$ higher electron thermal velocity $\rightarrow$ higher T_e

Electron Density from Absolute Intensity

Density Calculation:

$$n_e = N_{\text{total}} / (C \times E_{\text{total}} \times \sigma_T)$$

Where: - $N_{\text{total}} = \int \text{Thomson spectrum } d\lambda$ (total scattered photons) - C: Calibration factor from Raman ($\text{m}^{-3} \text{J}^{-1}$) - E_{total} : Total laser energy (J) - $\sigma_T = 7.94 \times 10^{-30} \text{ m}^2$ (Thomson differential cross-section at 90°)

Plasma Screening Parameter (α)

Definition:

$$\alpha = 1 / (k \times \lambda_D)$$

Where: - $k = (4\pi/\lambda_{\text{laser}}) \sin(\theta/2)$ (scattering wave vector, $\theta = 90^\circ$) - $\lambda_D = \sqrt{(\epsilon_0 k_B T_e / (n_e e^2))}$ (Debye length)

Relation to Density:

$$n_e = (\alpha^2 k^2 \epsilon_0 k_B T_e) / e^2$$

Physical Interpretation: - $\alpha \ll 1$: Incoherent scattering (individual electrons) - $\alpha \sim 1$: Transitional regime - $\alpha \gg 1$: Coherent/collective scattering (plasma waves)

Coherent Thomson Scattering Spectrum

Spectral Form Factor (Salpeter approximation):

$$S_e(k, \omega) = (1/\sqrt{\pi}) \times \exp(-x_e^2) / |1 + \alpha^2 W(x_e)|^2$$

Where: - $x_e = \Delta\omega / (k v_e)$ (normalized frequency shift) - $v_e = \sqrt{(2 k_B T_e / m_e)}$ (electron thermal velocity) - $W(x_e)$ = plasma dispersion function (includes Dawson integral) - $\Delta\omega = \omega_{\text{scattered}} - \omega_{\text{laser}}$ (frequency shift)

Key Features: - Central peak: free electron scattering - Satellites: ion acoustic waves (if $\alpha > 1$) - Asymmetry: collective effects

Fitting Methods

Method 1: Int.Calib.Gauss (Simplest)

Model: Gaussian approximation to Thomson spectrum

Fitted Parameters: - A: Amplitude - λ_0 : Center wavelength - FWHM: Full width at half maximum - b: Baseline offset

Gaussian Model:

$$I(\lambda) = A \times \exp(-4 \ln 2 \times ((\lambda - \lambda_0) / \text{FWHM})^2) + b$$

Calculations: 1. Fit Gaussian to spectrum → get FWHM, A 2. Calculate T_e from FWHM (see FWHM- T_e relation) 3. Calculate total counts: $N_{\text{total}} = \int I(\lambda) d\lambda \approx A \times \text{FWHM} \times \sqrt{\pi/(4\ln 2)}$ 4. Calculate $n_e = N_{\text{total}} / (C \times E_{\text{total}} \times \sigma_T)$

Pros: - □ Fast computation - □ Robust convergence - □ Good for initial estimates

Cons: - □ Assumes Gaussian shape (ignores collective effects) - □ Less accurate for $\alpha > 0.5$ - □ No α fitting

When to Use: Quick analysis, low-density plasmas ($\alpha < 0.3$), initial estimates

Method 2: Shape Calib. (Shape-Based)

Model: Full coherent Thomson scattering with plasma effects

Fitted Parameters: - T_e : Electron temperature (eV) - α : Plasma screening parameter - A: Amplitude (arbitrary normalization)

Objective Function:

$$\text{minimize } \sum [I_{\text{exp}}(\lambda) - A \times S_{\text{coherent}}(\lambda, T_e, \alpha)]^2$$

Where $S_{\text{coherent}}(\lambda, T_e, \alpha)$ is calculated from Salpeter theory.

Calculations: 1. Fit (T_e, α, A) to match spectral shape 2. Calculate n_e from α and T_e using dispersion relation 3. **Does NOT use** absolute intensity calibration

Pros: - □ Accounts for collective scattering effects - □ Independent verification of n_e - □ Sensitive to plasma physics

Cons: - □ n_e depends on α fit (not direct intensity measurement) - □ More sensitive to spectral shape errors - □ Can differ from intensity-based n_e

When to Use: Verification, consistency checks, high- α plasmas

Diagnostic Output:

NOTE - Shape Calib uses n_e from α (shape only), not intensity:
 n_e from α (used) = $2.45 \times 10^{17} \text{ m}^{-3}$
 n_e from intensity = $5.82 \times 10^{17} \text{ m}^{-3}$ (ratio: 2.38 x)

This shows how n_e from shape compares to what intensity calibration would give.

Method 3: Int.Calib.Full (Recommended)

Model: Full coherent Thomson scattering with absolute intensity

Fitted Parameters: - T_e : Electron temperature (eV) - α : Plasma screening parameter - A: Amplitude (physical units)

Two-Stage Process:

Stage 1: Shape Fitting

$$\text{minimize } \sum w(\lambda) \times [I_{\text{exp}}(\lambda) - A \times S_{\text{coherent}}(\lambda, T_e, \alpha)]^2$$

Where $w(\lambda)$ = Gaussian weights centered at 532 nm:

$$w(\lambda) = \exp(-((\lambda - 532) / \sigma)^2), \quad \sigma = 5 \text{ nm}$$

Why Weighted: Prioritizes central region where signal is strongest, prevents fits being pulled by noisy wings.

Stage 2: Intensity Calculation 1. Calculate total counts: $N_{\text{total}} = \int A \times S_{\text{coherent}}(\lambda, T_e, \alpha) d\lambda$ 2. Calculate $n_e = N_{\text{total}} / (C \times E_{\text{total}} \times \sigma_T)$ 3. **Independent of α** - uses direct photon counting

Pros: - ☐ Best absolute n_e accuracy (uses Raman calibration) - ☐ Accounts for collective effects in shape - ☐ Weighted fitting improves central peak fit - ☐ Two independent measurements (shape $\rightarrow T_e$, intensity $\rightarrow n_e$)

Cons: - ☐ Slowest method (~10-30 seconds) - ☐ Requires accurate Raman calibration - ☐ Most sensitive to calibration errors

When to Use: Final quantitative analysis, publication-quality results

Error Estimation (Monte Carlo Method)

All three fitting methods use **Monte Carlo sampling** to estimate realistic parameter uncertainties.

Traditional Method (Jacobian - Too Small)

$$\sigma_{\text{param}} = \sqrt{(\text{diag}(\sigma_{\text{residual}}^2 \times (J^T J)^{-1}))}$$

Problem: Only captures statistical noise, ignores model uncertainty $\rightarrow$ errors 10-100 $\times$ too small!

Monte Carlo Method (V9.9 - Realistic)

Algorithm:

1. **Perform best fit** $\rightarrow$ get best_ T_e , best_ n_e , best_SSR

2. **Define acceptance threshold:**

$$\text{SSR}_{\text{threshold}} = 1.5 \times \text{best_SSR}$$

(Adjust 1.5 for more/less conservative errors)

3. **Grid search parameter space:**

$$T_e_{\text{range}} = [0.7 \times \text{best_}T_e, 1.3 \times \text{best_}T_e] \text{ (25 points)}$$

$$n_e_{\text{range}} = [0.7 \times \text{best_}n_e, 1.3 \times \text{best_}n_e] \text{ (25 points)}$$

4. **For each (T_e_{test} , n_e_{test}):**

- Generate synthetic spectrum
- Calculate SSR_test
- If SSR_test < SSR_threshold $\rightarrow$ accept parameters

5. **Calculate errors:**

$$T_e_{\text{error}} = (\max(T_e_{\text{accepted}}) - \min(T_e_{\text{accepted}})) / 2$$

$$n_e_{\text{error}} = (\max(n_e_{\text{accepted}}) - \min(n_e_{\text{accepted}})) / 2$$

Physical Interpretation: - Error = range of parameters producing “visually indistinguishable” fits - Captures both statistical noise AND model uncertainty - Typically gives 15-20% errors (matches empirical observation)

Console Output:

Calculating realistic parameter errors via Monte Carlo sampling...

SSR threshold for acceptable fits: 3.51e+04 (best fit SSR = 2.34e+04)
Monte Carlo results: 189 parameter sets within acceptable SSR
Te range: 1.21 to 1.87 eV (error = 0.33 eV, 21.7%)
ne range: 1.45e+17 to 3.21e+17 m⁻³ (error = 8.80e+16 m⁻³, 19.4%)

Tuning Error Estimation

Line 1259 (Int.Calib.Full), 1407 (Shape Calib.), 1621 (Int.Calib.Gauss):

```
acceptable_ssr_threshold = best_resnorm * 1.5;
```

- **Increase multiplier** (e.g., 2.0) → larger errors (more conservative)
 - **Decrease multiplier** (e.g., 1.3) → smaller errors (tighter bounds)
 - **Recommended:** 1.3-1.7 depending on data quality
-

Workflow Guide

Standard Analysis Procedure

1. Data Preparation

- Ensure Raman and Thomson images are in same format (TIFF, PNG)
- Record acquisition parameters:
 - Laser shots
 - Laser energy per shot
 - Gas pressure and composition
- Obtain background images (laser off) if available

2. Initial Setup

```
% Edit RSTSGUI2.m lines 26-107
default_set = 2; % Choose your dataset

% Update file paths
p.raman_file = 'path/to/raman.tif';
p.thomson_file = 'path/to/thomson.tif';
p.raman_bg_file = 'path/to/raman_bg.tif'; % optional
p.thomson_bg_file = 'path/to/thomson_bg.tif'; % optional

% Set acquisition parameters
p.shots_raman = 300;
p.shots_thomson = 300;
p.energy_J_raman = 52e-3; % 52 mJ
p.energy_J_thomson = 52e-3;
```

3. ROI Adjustment

- **ROI yc:** Center of signal vertically (pixel row)
- **ROI wy:** Width of signal vertically (pixels)
- **Excl start-end:** Columns to exclude (laser line, typically 455-550)

Tips: - Look at raw images to identify signal region - Make ROI wide enough to capture all signal (typically 30-60 pixels) - Exclude enough around laser line to avoid saturation

4. Wavelength Calibration

Initial Calibration (if known):

```
p.a_initial = 0.010050; % nm/pixel  
p.b_initial = 526.7384; % nm offset
```

Optimization: - Click “**Recalib disp. from Raman**” - Uses peak matching between experimental and theoretical Raman - Updates a and b coefficients - Check that Raman peaks align in spectrum plot

Manual Fine-Tuning: - Adjust a and b fields directly - Click “**Recalc Raman Spectrum**” to update - Iterate until theoretical/experimental peaks align

5. Raman Analysis Verification

Check Raman spectrum plot: - **Good calibration:** Blue (experimental) and red (model) peaks aligned - **Calibration factor:** Should be $\sim 10^{-15}$ to 10^{-14} (order of magnitude) - **If mismatch:** Re-run calibration or adjust gas parameters

6. Thomson Fitting

Option A: Quick Analysis - Use initial Gaussian fit (displayed on file load) - Click “**Int.Calib.Gauss**” for Monte Carlo errors - Fastest, good for surveys

Option B: Accurate Analysis (Recommended) - Click “**Int.Calib.Full**” - Wait ~ 10 -30 seconds for weighted fit + Monte Carlo - Check console output for diagnostics - Best absolute accuracy

Option C: Verification - Click “**Shape Calib.**” - Compare ne from shape vs. intensity - Ratio should be 1 - $3\times$ typically - Large discrepancy ($>5\times$) indicates calibration issues

7. Result Validation

Check Fit Quality: - **R²:** Should be > 0.95 for good fits - **SSR:** Lower is better (compare between methods) - **Visual:** Fitted curve should match data well

Check Errors: - Should be 10-25% typically - Much smaller ($<5\%$) $\rightarrow$ suspicious, check MC sampling - Much larger ($>30\%$) $\rightarrow$ noisy data or poor fit

Cross-Validation: - Compare T_e between methods $\rightarrow$ should agree within errors - Compare ne from Int.Calib.Full vs. Shape Calib. $\rightarrow$ typically 1 - $3\times$ ratio

8. Parameter Sweeps

To analyze multiple files:

```
% In GUI file list (right panel)  
% 1. Click on file name  
% 2. Wait for automatic reanalysis  
% 3. Click fitting button of choice  
% 4. Save results  
  
% Or: loop programmatically  
files = dir('*.tif');  
for i = 1:length(files)  
    % Load file, fit, save results  
end
```

9. Data Export

CSV Export: - Click “**Save CSV**” - Creates file:
TS_<method>_Te<value>_ne<value>_SSR<value>.csv - Contains:
wavelength, experimental intensity, fitted intensity

Plot Export: - Click “**Save TS Spectrum**” (or Raman/Image buttons)
- Creates JPG files with embedded parameters in filename

Manual Export:

```
% Access from workspace after GUI runs:
handles = guidata(gcf);
Te = handles.edit_Te_manual.Value;
ne = handles.edit_ne_manual.Value;
wavelength = handles.plot_data.wavelength_nm;
spectrum = handles.current_ts_spectrum;
```

Parameter Tuning

Optimization Settings

Multi-start Fitting (Line 1196, 1363, 1562):

```
num_starts = handles.edit_multistarts.Value; % Default: 6
```

- **Purpose:** Avoid local minima in optimization
- **Typical:** 5-10 starts
- **More starts:** More robust, but slower
- **Fewer starts:** Faster, risk missing global minimum

GUI Control: “Multi” field (row 7)

Weighted Fitting (Int.Calib.Full)

Central Weighting (Line 1176):

```
central_wl = 532; % Center wavelength (nm)
weight_sigma = 5; % Decay width (nm)
```

Weight Function:

$$w(\lambda) = \exp(-((\lambda - 532)^2 / (2 \times 5^2)))$$

Effect: - $\sigma = 3$ nm: Very tight focus on center - $\sigma = 5$ nm: Moderate (default) - $\sigma = 8$ nm: Broader, includes more wings

When to Adjust: - **Narrow peaks** (cold plasma) → decrease σ - **Broad peaks** (hot plasma) → increase σ - **Noisy wings** → decrease σ (ignore wings)

Monte Carlo Sampling

Acceptance Threshold (Lines 1259, 1407, 1621):

```
acceptable_ssr_threshold = best_resnorm * 1.5;
```

Effect: - 1.3: Tighter errors (~10-15%) - 1.5: Moderate errors (~15-20%, default) - 2.0: Looser errors (~20-30%)

When to Adjust: - **High-quality data** (low noise) → 1.3 - **Noisy data** → 2.0 - **Publication** → 1.5 (standard)

Grid Resolution (Lines 1254, 1405, 1619):

```
Te_range = linspace(Te_fit * 0.7, Te_fit * 1.3, 25);  
ne_range = linspace(ne_fit * 0.7, ne_fit * 1.3, 25);
```

- **25 points:** Default (625 combinations)
 - **Increase** (e.g., 35): Finer resolution, slower
 - **Decrease** (e.g., 15): Faster, coarser errors
-

Gas Parameters

Pressure (Line 474):

```
handles.edit_pressure.Value = 101325; % Pa (1 atm)
```

Temperature (Line 479):

```
handles.edit_temp_K.Value = 293.15; % K (20°C)
```

Mixing Ratios (Lines 481, 483):

```
N2_fraction = 0.79; % 79% nitrogen  
O2_fraction = 0.21; % 21% oxygen
```

Important: N2 + O2 must equal 1.0 (automatically normalized)

Troubleshooting

Poor Fit Quality (Low R²)

Symptoms: R² < 0.90, large SSR, visual mismatch

Possible Causes: 1. **Incorrect wavelength calibration** - Solution: Run "Recalib disp. from Raman" - Verify Raman peaks align

2. **Wrong ROI/exclusion**
 - Solution: Adjust yc, wy, exclusion columns
 - Ensure signal fully captured, laser line excluded
 3. **Stray light/noise**
 - Solution: Use background subtraction
 - Enable outlier removal (RmOut checkbox)
 - Apply median filter (MedF, window=5)
 4. **Incorrect acquisition parameters**
 - Solution: Verify shots, energy, pressure, gas composition
 - Check that Raman calibration factor is reasonable ($\sim 10^{-15}$)
-

Very Small Errors (<5%)

Symptoms: Monte Carlo reports <5% errors

Possible Causes: 1. **Acceptance threshold too tight** - Solution: Increase multiplier (1.5 → 2.0)

2. **Few accepted parameter sets**
 - Check console: should be 100-300 sets
 - If <50 sets: widen search range or threshold
 3. **Perfect fit** (unusual)
 - May be real if data quality is exceptional
 - Verify by visual inspection
-

Large Errors (>30%)

Symptoms: Monte Carlo reports >30% errors

Possible Causes: 1. **Noisy data** - Solution: Average more shots - Use background subtraction - Enable filters

2. **Poor fit convergence**
 - Increase multi-start attempts
 - Check initial parameter guesses
 3. **Model mismatch**
 - Check α value (should be 0.1-2.0 typically)
 - Verify plasma assumptions valid
-

Density Discrepancies Between Methods

Symptoms: Int.Calib.Full and Shape Calib. give very different ne

Expected: Ratio of 1-3 $\times$ is normal

If Ratio > 5 $\times$: 1. **Check Raman calibration** - Verify gas pressure, temperature, composition - Re-run "Recalib disp. from Raman"

2. **Check Thomson acquisition parameters**
 - Verify shots, energy are correct
 - Ensure no saturation in images
3. **Check spectral shape**
 - If α is very high (>2), collective effects may be strong
 - Shape-based ne may be less reliable

Recommendation: Trust Int.Calib.Full for absolute ne (uses direct intensity measurement)

Fit Converges to Wrong Shape

Symptoms: Fitted spectrum visibly misaligned with data

Possible Causes: 1. **Local minimum in optimization** - Solution: Increase multi-start attempts (6 $\rightarrow$ 12)

2. **Bad initial guess**
 - Manually set Te, ne closer to expected values
 - Click "Recalc TS" to update
 3. **Wavelength calibration off**
 - Use pixel shift controls (< / >) to align
 - Re-calibrate from Raman
-

GUI Crashes or Errors

Common Issues:

1. **"File not found"**
 - Check file paths in lines 26-107
 - Use absolute paths
 2. **"Undefined function"**
 - Missing toolbox (Optimization, Image Processing)
 - Install required toolboxes
 3. **"Out of memory"**
 - Reduce image size
 - Close other MATLAB processes
 4. **"Invalid parameter"**
 - Check that N2 + O2 = 1.0
 - Verify all fields have valid numbers
-

Best Practices

Data Acquisition

- ☐ Use consistent laser energy (minimize shot-to-shot variation)
- ☐ Average 100-500 shots for good statistics
- ☐ Acquire background with laser off (same integration time)
- ☐ Use reference Raman scattering from known gas at known pressure
- ☐ Minimize stray light (proper baffling, alignment)

Analysis Workflow

- ☐ Start with Int.Calib.Gauss for quick initial estimates
- ☐ Use Int.Calib.Full for final quantitative results
- ☐ Verify with Shape Calib. for consistency
- ☐ Check that $R^2 > 0.95$ for all methods
- ☐ Compare errors across methods (should be similar)

Parameter Selection

- ☐ Use multi-start = 6-10 for robust fitting
- ☐ Use acceptance threshold = 1.5 for standard errors
- ☐ Adjust weighted fitting σ based on spectral width
- ☐ Re-calibrate wavelength if peaks don't align

Reporting Results

- ☐ Report method used (Int.Calib.Full recommended)
 - ☐ Include Monte Carlo errors (± 15 -20% typical)
 - ☐ State acquisition conditions (shots, energy, gas)
 - ☐ Report fit quality (R^2 , SSR)
 - ☐ Show fitted spectrum vs. data (visual verification)
-

Advanced Features

Custom Fitting Functions

To modify fitting behavior, edit these functions:

coherent_fitter_direct (Line 1487): - Calculates coherent Thomson spectrum - Modify to change shape model

calculate_coherent_ts (Line 1959): - Core Salpeter formula - Includes plasma dispersion function

intensity_calibration_callback (Line 1163): - Main fitting routine for Int.Calib.Full - Modify weighting, optimization

Batch Processing

Process multiple files programmatically:

```
% Define file list
files = {
    'file1.tif',
    'file2.tif',
    'file3.tif'
};
```

```

% Initialize results storage
results = struct('Te', [], 'ne', [], 'Te_err', [], 'ne_err', []);

% Loop through files
for i = 1:length(files)
    % Set Thomson file
    params.thomson_file = files{i};

    % Run analysis
    [res, data] = run_full_analysis(params);

    % Store results
    results.Te(i) = res.corrected.Te_eV;
    results.ne(i) = res.corrected.ne_m3;
    results.Te_err(i) = res.deconv_results.Te_eV_error;
    results.ne_err(i) = res.deconv_results.ne_error;
end

% Plot results
errorbar(1:length(files), results.Te, results.Te_err);
xlabel('File Index');
ylabel('Te (eV)');

```

Exporting Custom Data

Access internal data structures:

```

% After GUI is open
handles = guidata(gcf);

% Get all data
wavelength = handles.plot_data.wavelength_nm;
spectrum = handles.current_ts_spectrum;
fitted = get(handles.h_coherent_fit, 'YData');
Te = handles.edit_Te_manual.Value;
ne = handles.edit_ne_manual.Value;
alpha = handles.edit_alpha.Value;

% Export
save('my_analysis.mat', 'wavelength', 'spectrum', 'fitted', 'Te',
    'ne', 'alpha');

```

References

Theory

1. **Sheffield, J.** (1975). *Plasma Scattering of Electromagnetic Radiation*. Academic Press.
 - Comprehensive treatment of Thomson scattering theory
2. **Salpeter, E.E.** (1960). "Electron Density Fluctuations in a Plasma", *Phys. Rev.* **120**, 1528.
 - Original plasma dispersion function formulation
3. **Hutchinson, I.H.** (2002). *Principles of Plasma Diagnostics* (2nd ed.). Cambridge University Press.
 - Modern plasma diagnostic techniques

Experimental

4. **Muraoka, K. & Maeda, M.** (1991). "Laser-Aided Diagnostics of Plasmas and Gases", *IOP Publishing*.
 - Practical laser diagnostic methods
5. **Evans, D.E. & Katzenstein, J.** (1969). "Laser light scattering in

laboratory plasmas”, *Rep. Prog. Phys.* **32**, 207.
◦ Classic review of Thomson scattering diagnostics

Raman Scattering

6. **Long, D.A.** (2002). *The Raman Effect: A Unified Treatment of the Theory of Raman Scattering*. Wiley.
◦ Comprehensive Raman theory and applications

Error Analysis

7. **Press, W.H. et al.** (2007). *Numerical Recipes* (3rd ed.). Cambridge University Press.
◦ Monte Carlo methods and parameter estimation
-

Version History

V9.9 (October 2024) - RSTSGUI2

- Added weighted fitting prioritizing 532 nm central region
- Implemented Monte Carlo error estimation for all methods
- Fixed initial Gaussian fit ne calculation bug
- Added diagnostic output for Shape vs. Intensity ne comparison
- Unified error methodology across all fitting callbacks

V9.8 (Prior)

- Automatic Te, ne, alpha field updates after Gaussian fit
- Editable gamma field for duty cycle
- Multi-start optimization for Raman calibration
- Status message display

Earlier Versions

- Basic GUI framework
 - Gaussian and coherent fitting
 - Raman calibration
 - ROI and exclusion controls
-

Support

For issues, questions, or contributions:

GitHub: <https://github.com/ShuraBura/Raman-Thomson-Scattering-GUI>
Branch: claude/matlab-support-011CUNPs1aDJHFFbyb5mLPrh

Reporting Bugs: 1. Provide MATLAB version 2. Describe steps to reproduce 3. Include error messages 4. Attach sample data if possible

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Monte Carlo error estimation and weighted fitting improvements
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Version:** RSTSGUI2 V9.9